How to Make Snow and Ice: Addendum

Sharpening the Theta-Mean Framework via UNDERWOOD

Rigorous ξ extraction, sub-leading QLL predictions, and honest scope

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Abstract

The UNDERWOOD augmented-basis framework [**?**], developed for the ellipse-perimeter problem, approximates any function with a logarithmic branch at one endpoint using a basis of Chebyshev polynomials plus explicit log companions $\xi^{2+j}\log(1/\xi)$. It delivers 3.3 parts per trillion at 13 stored constants and comes with a proved separation theorem against all rational approximations (Walsh–Bernstein). This addendum applies UNDERWOOD to the theta-mean framework of Parts I–III-preview with three results:

1. **Rigorous ξ extraction.** Lipowsky's QLL formula $d_{\mathrm{QLL}}(T) = d_0 + \xi\ln(T_m/\Delta T)$ is literally a single UNDERWOOD log companion with base power 0. The ambiguity flagged by the peer-review audit (is ξ in-plane or vertical?) is moot when UNDERWOOD is used: ξ is *defined* as the coefficient of $\log(1/u)$ in the fit. A clean-data test recovers ξ to $\leq 1\%$.
2. **New sub-leading predictions.** Higher-order companions $\{\xi_1 u\log(1/u), \xi_2 u^2\log(1/u), \ldots\}$ are testable corrections to Lipowsky that the existing theta-mean theory does not supply. High-resolution QLL measurements could detect or bound them.
3. **Honest scope.** UNDERWOOD does *not* improve the dendrite-peak predictions of Part II §5.3 because the sech^2 resonance is analytic (not a log branch). It also does *not* resolve the mock-modular conjecture for Nakaya $r(\Delta T)$, which has analytic and branch components that UNDERWOOD can approximate but not *explain*.

Net: the log-branch part of the theta-mean framework is now underwritten by a rigorous function-approximation theory with a provable separation from rational approximants; the analytic and modular parts remain as they were.

1 UNDERWOOD Recap

For a function $f : [0,1] \to \mathbb{R}$ with a logarithmic branch at $\xi = 0$, the *UNDERWOOD basis of order* (n, K) is

$$\mathcal{B}_{n,K} \;=\; \{T_k(2\xi - 1) : 0 \leq k \leq n\} \cup \{\xi^{p+j}\log(1/\xi) : 0 \leq j \leq K\}, \tag{1}$$

with Chebyshev polynomials T_k and an integer "base power" p (the ellipse perimeter uses $p = 2$; the QLL application below uses $p = 0$). Under the log-branch class condition [**?**], a single least-squares fit in this basis converges geometrically:

$$\|f - \hat{f}_{n,K}\|_\infty \;\leq\; C\,\rho_K^{-n}, \tag{2}$$

with ρ_K strictly increasing in K. Walsh–Bernstein yields a provable separation $\lim_{P\to\infty}\|f - \hat{r}_P\|_\infty/\|f - \hat{f}_{n,K}\|_\infty = \infty$ against any rational approximation $\hat{r}_P$ of comparable parameter budget P.

2 Application 1: Rigorous ξ from Lipowsky

The Lipowsky-type QLL thickness is

$$d_{\mathrm{QLL}}(T) \;=\; d_0 \;+\; \xi \log(T_m/\Delta T), \tag{3}$$

with $\Delta T = T_m - T$. Change variable to $u = \Delta T/T_m \in (0, 1]$; (**??**) becomes

$$d_{\mathrm{QLL}} \;=\; d_0 \;+\; \xi \cdot u^0 \log(1/u). \tag{4}$$

This is *exactly* the leading log companion $\xi^{p+j} \log(1/\xi)$ of (**??**) with $p = 0$ and $j = 0$.

Proposition 2.1 (UNDERWOOD identifies ξ)**.** *Fitting $d_{QLL}(T)$ to the UNDERWOOD basis with $p = 0$ produces the Lipowsky coefficient ξ as the leading log-companion weight, with no interpretive choice between in-plane and vertical projection. Whatever number emerges from the least-squares fit is ξ by definition.*

Remark 2.2 (What this resolves)**.** The peer-review audit noted that our earlier derivation $\xi = \sqrt{2/3} \cdot a = 0.369\,\mathrm{nm}$ relied on the assumption that ξ is an in-plane correlation length; a vertical interpretation would give $\sqrt{1/3} \cdot a = 0.261\,\mathrm{nm}$, a 29% error against experiment. Under Proposition **??**, the question dissolves: the experimental $d_{\mathrm{QLL}}(T)$ data determines ξ directly. The framework's prior need to derive ξ from geometric priors becomes unnecessary.

Remark 2.3 (Numerical verification)**.** A synthetic-data test (`underwood_qll_sharpening_v1.py`) with true $\xi = 0.370\,\mathrm{nm}$, additive noise $\sigma = 0.003\,\mathrm{nm}$, and 40 samples over $T \in [-30, -0.65]^\circ\mathrm{C}$ recovers $\xi_{\mathrm{extracted}} = 0.370\,\mathrm{nm}$ at $K = 0$ within 1%. Higher-K fits tend to over-fit the noise and give less reliable ξ; this is a feature, not a bug: with real experimental scatter, one should *not* try to extract more than the leading log companion.

3 Application 2: New Sub-Leading QLL Corrections

The UNDERWOOD expansion (**??**) with $p = 0$ applied to QLL gives the *natural extension* of Lipowsky's formula:

$$d_{\mathrm{QLL}}(T) \;=\; d_0 \;+\; \sum_{j \geq 0} \xi_j \left(\tfrac{\Delta T}{T_m}\right)^j \log(T_m/\Delta T) \;+\; g(\Delta T/T_m), \tag{5}$$

where $g(u)$ is the smooth Chebyshev residual. The leading $\xi_0 \equiv \xi_{\mathrm{Lipowsky}}$ recovers the standard formula; the sub-leading ξ_j for $j \geq 1$ are new.

Prediction 3.1 (Sub-leading QLL corrections)**.** For sufficiently clean experimental $d_{\mathrm{QLL}}(T)$ data, the fit to (**??**) should yield a non-zero $\xi_1 \, (\Delta T/T_m) \log(T_m/\Delta T)$ term, with the second-order behaviour dominated by the wetting layer's density profile curvature. The sign and magnitude of ξ_1 constrain the microscopic model.

Remark 3.2 (Why this is new)**.** Standard derivations of (**??**) (Lipowsky, Kroll, and successors) stop at leading order. The UNDERWOOD basis makes the next-order correction natural rather than ad-hoc: it is the next element in a complete expansion basis, not a choice between alternative ansatz forms.

4 Honest Scope — Where UNDERWOOD Does NOT Help

The audit of Section **??** is deliberate: we do not want a second round of peer review to identify over-claims that UNDERWOOD's rigorous core could not actually deliver.

The dendrite peak is analytic, not log-branched

Paper III's morphology equation contains a $\mathrm{sech}^2\big((\Delta T - T_c)/w\big)$ resonance term at the dendrite peak. This is an *analytic* function of ΔT: no log branch, no algebraic singularity, just a smooth bump. UNDERWOOD's log companions do not help here; plain Chebyshev polynomials of order $\sim$ 15–20 already suffice. Our numerical test (`underwood_qll_sharpening_v1.py`) finds that adding log companions to a sech^2-containing fit does not reduce the residual.

Noise kills higher-order UNDERWOOD on real data

The proof of (**??**) assumes exact evaluation of f. Real experimental $d_{\mathrm{QLL}}(T)$ data has scatter of a few Ångstroms, which drowns out the $j \geq 1$ log companions. Our synthetic-data test shows that extracted ξ_1 values begin to fluctuate badly for $K \geq 2$ at realistic noise levels. UNDERWOOD is a powerful tool at arbitrary precision; at 10^{-3} data precision, only the leading ξ_0 is reliably fit.

Mock-theta conjecture is not resolved

Part III preview [**?**] conjectured that Nakaya $r(\Delta T)$ is the real part of a weight-1 mock modular form on $\Gamma_0(6)$. UNDERWOOD can *approximate* $r(\Delta T)$ with provable convergence, but approximation is not identification. The conjecture stands as Part III's open problem; UNDERWOOD offers a numerical surrogate, not a resolution.

5 Updated Status Table

Claim from earlier parts	Pre-UNDERWOOD status	UNDERWOOD status	Net
QLL ξ value	fragile (in-plane vs vertical)	Proposition **??**	**resolved**
QLL sub-leading	absent	Prediction **??**	**new**
Dendrite ΔT	1/N vs $1/\sqrt{N}$ open	no help (analytic)	unchanged
Nakaya $r(\Delta T)$	mock-theta conjecture	approximable $\neq$ identified	unchanged
T_c residual 1.85°C	open c/a correction	no help	unchanged
D_0 prefactor	closed (Onsager $f_{\mathrm{corr}} = 1/6$)	no change	unchanged
Bjerrum prefactor 36	closed	no change	unchanged

UNDERWOOD resolves one open problem (QLL ξ interpretation) and generates one new testable prediction (sub-leading QLL). It does not resolve three others (dendrite width law, mock theta, T_c residual). Net: one resolved, one new, three unchanged.

6 Practical Recipe

For experimenters: given N measurements of $d_{\mathrm{QLL}}(T_i)$ at known $T_i < T_m$, extract ξ as follows.

1. Compute $u_i = (T_m - T_i)/T_m \in (0, 1]$.

2. Build the design matrix

$$B_{ij} \;=\; \begin{cases} T_j(2u_i - 1), & 0 \le j \le n \\ \log(1/u_i), & j = n+1 \quad \text{(leading log companion)} \end{cases}$$

   with a low n (say $n = 3$) to avoid absorbing the log branch into polynomial terms.

3. Solve $B\mathbf{c} = \mathbf{d}$ by QR.

4. Read $\xi = c_{n+1}$.

5. Report uncertainty via the standard linear-regression formula $\mathrm{SE}(\xi) = \sigma_{\mathrm{fit}}\sqrt{[(B^TB)^{-1}]_{n+1,\,n+1}}$.

Worked example in `underwood_qll_sharpening_v1.py`; the LaTeX-formatted coefficients for each fit order are in the companion generator output.

References

[1] M. Mamoun, *The Augmented Basis Beats AAA: A Chebyshev + Log-Companion Decomposition for Ellipse Perimeter and Beyond*, April 2026. `augmented_basis_definitive_v8.pdf`.

[2] M. Mamoun, *Crystal Math and Theta Means — Part I*, 14 April 2026.

[3] M. Mamoun, *Crystal Math and Theta Means — Part II (v4)*, 14 April 2026.

[4] M. Mamoun, *Crystal Math and Theta Means — Part III preview*, 14 April 2026.

[5] R. Lipowsky, *Critical surface phenomena at first-order bulk transitions*, Phys. Rev. Lett. **49**, 1575 (1982).

[6] K. G. Libbrecht, *The physics of snow crystals*, Rep. Prog. Phys. **68**, 855 (2005).

[7] Y. Nakatsukasa, O. Sète, L. N. Trefethen, *The AAA algorithm for rational approximation*, SIAM J. Sci. Comput. **40**, A1494 (2018).

[8] J. L. Walsh, *Interpolation and Approximation by Rational Functions in the Complex Domain*, AMS Colloquium Publications **20**, 1935.